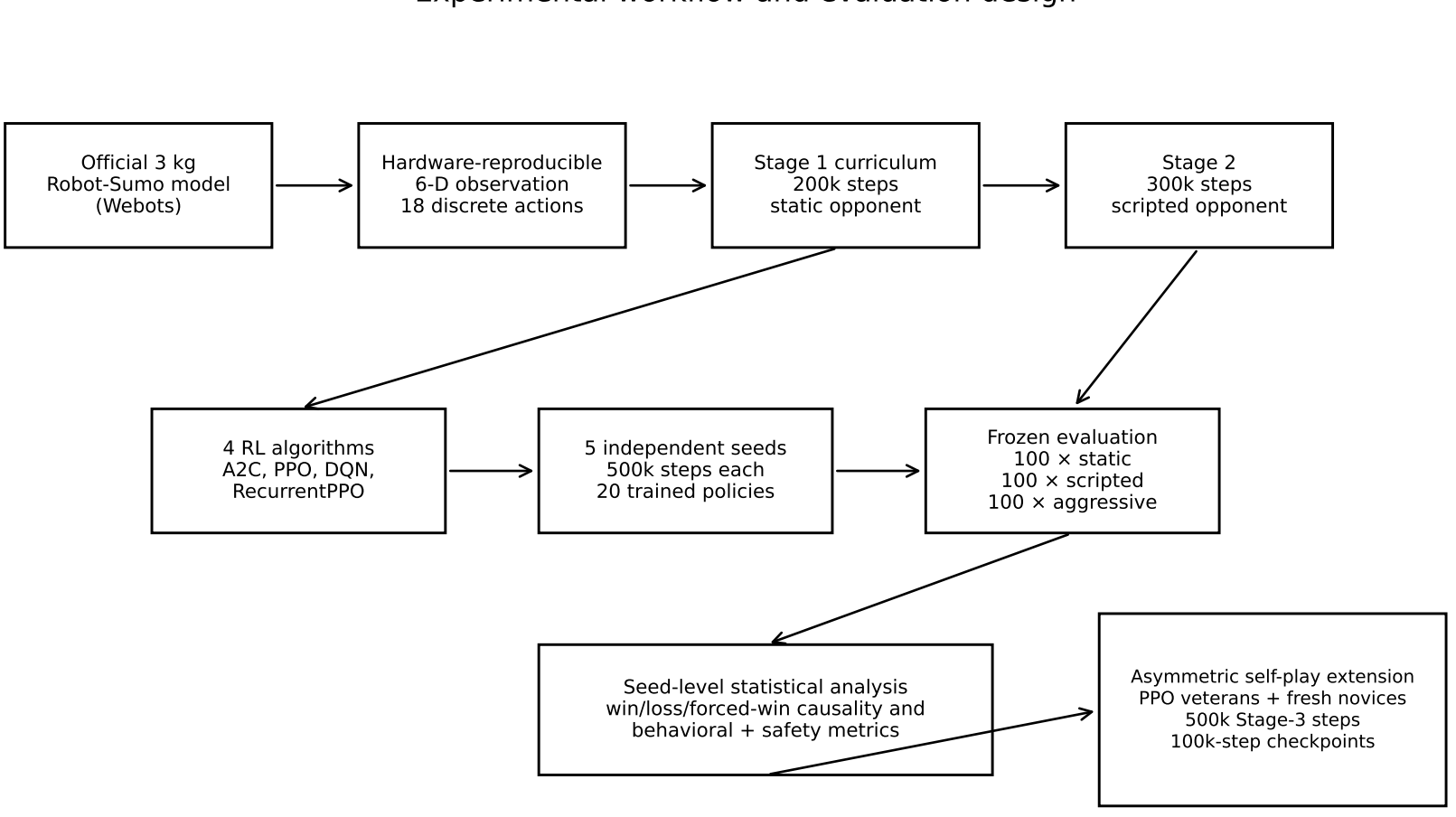
Experimental workflow and evaluation design
Official 3 kg
Robot-Sumo model
(Webots)
Hardware-reproducible
6-D observation
18 discrete actions
Stage 1 curriculum
200k steps
static opponent
Stage 2
300k steps
scripted opponent
4 RL algorithms
A2C, PPO, DQN,
RecurrentPPO
5 independent seeds
500k steps each
20 trained policies
Frozen evaluation
100 × static
100 × scripted
100 × aggressive
Seed-level statistical analysis
win/loss/forced-win causality and
behavioral + safety metrics
Asymmetric self-play extension
PPO veterans + fresh novices
500k Stage-3 steps
100k-step checkpoints